The Stark problem in a Debye plasma: a coupled-channel complex-scaling map of field–screening competition

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A hydrogen atom in a Debye plasma subjected to a static electric field supports resonances shaped by two competing decay routes: ionization by the field and dissolution by screening. We compute them with a coupled-channel complex-scaling method whose radial channels carry the exact orthonormal Laguerre basis of the unperturbed problem, so the non-separable system stays below 10^3 degrees of freedom while reproducing the classical pure-Stark benchmark to ten digits, with every assembly verified at run time against exact limits. On this basis we map the ground-state resonance on an 85-point grid of the (F, λ) plane — to our knowledge the most extensive and finest-grained such map reported to date — and follow the $n = 2$, $m = 0$ Stark doublet in the screening parameter. Screening amplifies field ionization exponentially: at $F = 0.02$ a.u. the width grows by more than eight orders of magnitude across the screening range, against a single order at $F = 0.10$, quantifying a trend first observed at isolated parameter points by Yu and Ho. The operational frontier $\Gamma = |E_r|$ recedes to weaker screening as the field grows, the perturbative Stark regime shrinks with screening until the width, not the shift, dominates the field response, and the uphill Stark component is pushed across $E = 0$ — the ionization threshold of the field-free problem — continuing smoothly as a positive-energy resonance.

I. INTRODUCTION

The Stark effect in hydrogen and the screened Coulomb (Yukawa–Debye) potential are two classical single-parameter problems whose resonance structure is known to high precision: the pure Stark resonances from asymptotic and complex-dilation theory [1–5], and the screened-Coulomb bound states and critical screenings from decades of plasma-physics work [6–9]. Their combination — an atom in a plasma subjected to an electric field — is qualitatively richer: the field tilts the potential, screening lifts and delocalizes the bound states, and the two decay mechanisms compete. The combined problem is non-separable (screening breaks the parabolic separability of the Stark problem), and its record is substantial but scattered: Yu and Ho computed ground-state positions and widths by complex-coordinate rotation over a range of Debye lengths and fields [10]; Paul and Ho treated the bound levels, including the $n = 2$ splitting, at the level of energies [11]; Lin and Ho extended the complex-scaled treatment to ground and selected excited states with a Lagrange mesh [12]; bound-state studies at weak fields continue for generalized screening models [13, 14]; and for the exponential-cosine screened potential of quantum plasmas, Stark resonance energies and widths have been obtained by complex scaling [15]. What these works tabulate are selected parameter combinations. What is still missing is a *map*: dense, uniform two-parameter coverage of positions and widths under a single error protocol, from which the global structure of the competition — amplification laws, critical curves, the fate of excited manifolds — can be read directly. To our knowledge, the present 85-point grid with uniform uncertainty control is the most extensive such coverage reported for the Debye–Hückel potential.

Two methodological choices make the map cheap enough to be routine. First, instead of a two-dimensional mesh we exploit the dipole selection rule: the field couples only adjacent angular channels, so the problem is a block-tridiagonal radial system in which each channel carries the *exact* orthonormal Laguerre basis of the unperturbed radial problem [16], and the whole matrix is complex-scaled analytically. Convergence in the channel cutoff is geometric, and the classical pure-Stark benchmark is reproduced to ten digits with fewer than 10^3 degrees of freedom. Second, every assembly is verified at run time against exact identities. This discipline earns its keep: during development it exposed, within seconds, a factor-of-two kinetic-energy defect that had produced a convergent, rotation-stable spurious state at a plausible energy — precisely the failure mode to which complex-rotation width calculations are known to be sensitive.

II. METHOD

In the $m = 0$ sector the Hamiltonian of a hydrogen-like atom in a Debye plasma and a static field $F\hat{z}$ (atomic units),

$$H = -\tfrac{1}{2}\nabla^2 - \frac{Ze^{-\lambda r}}{r} + Fz, \tag{1}$$

acts on channel functions $u_\ell(r)$ coupled only by the dipole matrix elements of $z = r\cos\theta$: $\langle \ell+1, 0|\cos\theta|\ell, 0\rangle = (\ell+1)/\sqrt{(2\ell+1)(2\ell+3)}$. Each channel is expanded in the orthonormal generalized-Laguerre radial functions of Ref. [16] with a *common* scale β and a shared quadrature grid across channels, so that the inter-channel dipole

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blocks are exact contractions. Production settings: $\beta = 0.35$, $n_b = 70$ functions per channel, $\ell_{\max} = 14$ (1050 degrees of freedom), and a 3000-point Gauss–Legendre grid on $x \in [0, 760]$, $x = 2\beta r$; the anchors of Table I use $(\ell_{\max}, n_b) = (12, 60)$ and $(16, 80)$. Because the basis is orthonormal (the overlap identity is certified to 1.3×10^{-12} on every constructed basis), the discretization yields a standard — not generalized — eigenproblem for a complex-symmetric block-tridiagonal matrix.

Complex scaling $r \to re^{i\theta_c}$ acts analytically on every block: kinetic terms scale by $e^{-2i\theta_c}$, the Coulomb factor by $e^{-i\theta_c}$ with the screening exponential evaluated at complex argument, and the linear Stark coupling by $e^{+i\theta_c}$ [17–21]. For the unbounded Stark term this is justified not by dilation analyticity of Balslev–Combes type but by the Herbst–Simon theory of the scaled Stark operator, whose essential spectrum is empty and whose resonance eigenvalues are locally analytic in the parameters [2, 22]. The oscillatory screened blocks $e^{-\lambda r\cos\theta_c}e^{-i\lambda r\sin\theta_c}$ are the quadrature stress point; at the largest $\lambda\theta_c$ used ($\lambda = 0.8$, $\theta_c = 0.28$) doubling the grid changes no screened-potential or dipole matrix element by more than 10^{-10}.

Resonances are identified by parametric continuation plus rotation stability. The search window for each new λ is predicted as the exact $F = 0$ Yukawa ground level (one symmetric eigensolve) plus the previous point's complex field shift; on the computed grids the predictor lands within $\sim10^{-3}$ of the eigenvalue while the nearest competing (rotated-continuum) eigenvalue is an order of magnitude farther, and the value is accepted only if the rotations $\theta_c = 0.18$ and 0.28 agree (retry at 0.30/0.42), the modulus of their difference being recorded as the spread. Two-angle agreement alone is a necessary condition, so for the two least favorable points we computed full θ_c-trajectories over $[0.15, 0.45]$: the broad ground state at $F = 0.02$, $\lambda = 0.8$ is stationary to 2×10^{-9} across the entire range, and the positive-energy $n = 2$ component at $\lambda = 0.30$ has a true plateau for $\theta_c \geq 0.25$ with residual variation 4×10^{-5} at the production angles — consistent with the spreads reported below.

Verification: the hydrogen limit of the assembled system is checked against the exact levels $-1/2n^2$ (deviation $< 10^{-10}$ on the lowest levels) before every production run — a failed check now aborts the run — and the two one-parameter limits are anchored independently in Sec. III. Quoted uncertainties are spreads and inter-grid drifts, i.e. error *estimates* calibrated against the benchmarks, not rigorous bounds. A remark on solver scale: at these dimensions dense complex diagonalization computes each map point in seconds, so contour-integral eigensolvers, useful at mesh scales, are unnecessary here.

TABLE I. Pure-Stark ($\lambda = 0$) hydrogen ground-state resonances, $E = E_r - i\Gamma/2$ (a.u.). Drift = distance between the two convergence grids.

F	E_r	$\Gamma/2$	drift
0.04	−0.5037715909	0.0000019463	2×10^{-11}
0.08	−0.5175606169	0.0022698288	3×10^{-10}
0.10	−0.5274181749	0.0072690567	2×10^{-9}

III. RESULTS

A. Validation anchors

Table I collects the pure-Stark ground-state anchors. Each entry is computed on two independent discretizations and two rotation angles; the fine-grid value is quoted and the inter-grid drift estimates the truncation error. At $F = 0.10$ the result agrees with the classical complex-dilation value [3–5] on all quoted digits. In the opposite limit ($F = 0$, $\lambda = 0.25$, $\ell = 1$ block, $n_b = 220$) the method gives $E = 0.003064450 - 0.002184741i$, within 2.4×10^{-7} of the triple-verified benchmark of the one-dimensional engine [16] — the two edges of the parameter plane are anchored independently.

B. The ground-state (F, λ) map

Figure 1 shows the ground-state map: E_r, the field-induced displacement $E_r - E_b^{F=0}$ relative to the screened bound level, and Γ on the full grid $F \in \{0.02, \ldots, 0.10\} \times \lambda \in \{0, 0.05, \ldots, 0.80\}$; Table II lists a subset. All 85 points are θ_c-stable, with spreads below 7×10^{-6} at the broadest corner and below 10^{-7} over 73% of the grid.

Three features organize the map.

Screening-amplified field ionization. That widths grow when screening strengthens at fixed field was observed at individual parameter points already in Refs. [10, 12]; the map quantifies the law and its dramatic field dependence. From $\lambda = 0$ to 0.8 the amplification factor is $\sim10^4$ at $F = 0.04$, 135 at $F = 0.06$, 22 at $F = 0.08$, and 9 at $F = 0.10$; at $F = 0.02$ the $\lambda = 0$ width lies below the 10^{-10} resolution while the $\lambda = 0.8$ width is 1.3×10^{-2} — more than eight orders of magnitude. The asymmetry has a transparent quasistatic origin: screening lifts the level, and the tunneling exponent $\exp[-\frac{2}{3F}(2|E_r|)^{3/2}]$ [1] reacts exponentially. This argument applies in the sub-barrier regime $F \lesssim E_r^2/4$, which at $F = 0.02$ means $\lambda \lesssim 0.3$; there it is quantitative — between $\lambda = 0.2$ and 0.3 the exponent predicts a growth of 196 against a measured 96 — while beyond it the state is over-barrier and the exponential serves only as a heuristic for the trend. The practical reading for plasma spectroscopy: a field that is spectroscopically harmless for the unscreened atom ($\Gamma < 10^{-10}$) can dominate the line width entirely at strong screening.

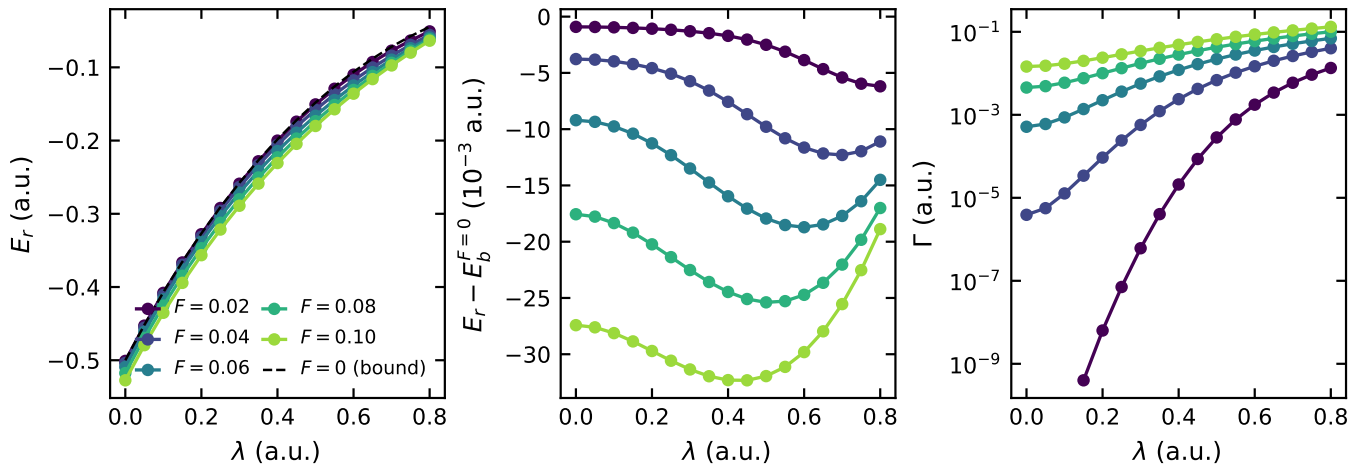


FIG. 1. Ground-state resonance across the (F, λ) plane. Left: position E_r; the dashed line is the $F = 0$ screened bound state. Center: field-induced displacement $E_r - E_b^{F=0}$. Right: width Γ (log scale; points are shown only where Γ exceeds ten times the θ_c-stability spread).

TABLE II. Ground-state resonance parameters (a.u.) at selected screenings. Full grids ($\Delta\lambda = 0.05$) are provided as machine-readable data. Widths marked $< 10^{-10}$ are below the certified resolution.

F	λ	E_r	Γ
0.02	0.0	-0.500909	$< 10^{-10}$
	0.4	-0.200084	2.094×10^{-5}
	0.8	-0.050898	1.344×10^{-2}
0.04	0.0	-0.503772	3.893×10^{-6}
	0.4	-0.205946	2.381×10^{-3}
	0.8	-0.055804	4.018×10^{-2}
0.06	0.0	-0.509204	5.151×10^{-4}
	0.4	-0.214339	1.209×10^{-2}
	0.8	-0.059208	6.970×10^{-2}
0.08	0.0	-0.517561	4.540×10^{-3}
	0.4	-0.222844	2.827×10^{-2}
	0.8	-0.061712	1.004×10^{-1}
0.10	0.0	-0.527418	1.454×10^{-2}
	0.4	-0.230677	4.870×10^{-2}
	0.8	-0.063587	1.316×10^{-1}

Dissolution frontier. As an operational criterion for the loss of spectroscopic identity we take $\Gamma = |E_r|$; any $O(1)$ ratio gives a similar curve. It is crossed inside the computed range only for $F \geq 0.06$: at $\lambda^* \approx 0.78$, 0.73, and 0.68 (± 0.02, linear interpolation on the $\Delta\lambda = 0.05$ grid) for $F = 0.06$, 0.08, and 0.10. The frontier moves toward weaker screening as the field grows, delimiting the region of the (F, λ) plane where a plasma-embedded atom retains a resolvable line.

Shrinking perturbative regime. The center panel of Fig. 1 isolates the field's contribution to the position. At $\lambda = 0$ the displacements are perturbative: -0.909×10^{-3} at $F = 0.02$ versus $-\frac{9}{4}F^2 = -0.9 \times 10^{-3}$, and at $F = 0.10$ the measured -27.4×10^{-3} is bracketed by the second- and fourth-order asymptotic estimates. Screening destroys this hierarchy rapidly. At $F = 0.02$ the displacement deepens by a factor 6.8 between $\lambda = 0$ and 0.8 — but this is *not* a growing polarizability in the quadratic sense: halving the field to $F = 0.01$ at $\lambda = 0.8$ reduces the displacement by only a factor 2.24 (quadratic response would give 4), so even the weakest fields considered are beyond the perturbative regime at strong screening. At strong field the displacement is non-monotonic in λ, peaking at -32.3×10^{-3} near $\lambda \approx 0.45$ for $F = 0.10$ and retreating to -18.9×10^{-3} at $\lambda = 0.8$; quadrupling the field from $F = 0.02$ to 0.04 at $\lambda = 0.8$ increases it by only 1.8×. Screening shrinks the perturbative Stark regime, and beyond it the dominant field response is the width, not the shift.

C. The $n = 2$ doublet across $E = 0$

In the $m = 0$ sector the $n = 2$ level carries exactly two components, labeled by the parabolic quantum numbers $k = \pm 1$ of the $\lambda = 0$ limit; since screening destroys parabolic separability, k is used throughout as a continuation label of the pole trajectory, not as a quantum number. At $F = 0.02$ — five times the classical ionization field of $n = 2$ — both components are broad resonances already at $\lambda = 0$: $E = -0.206682 - i\,0.030393$ ($k = -1$, downhill) and $-0.088984 - i\,0.015446$ ($k = +1$, uphill), the downhill component ionizing twice as fast and both positions deviating from second-order perturbation theory by $+0.012$ and $+0.010$. Figure 2 follows the doublet in λ up to 0.30 (θ_c-spreads below 10^{-5} except the two broadest points, worst 1×10^{-4}). Screening lifts both components at nearly the same initial rate (the slope $dE/d\lambda|_0 = \langle e^{-\lambda r} \rangle \to 1$ is state-independent), so the

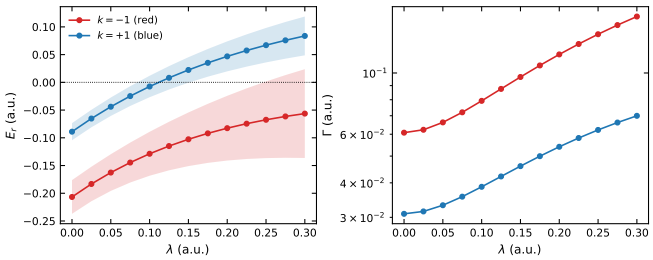


FIG. 2. The $n = 2$, $m = 0$ Stark doublet at $F = 0.02$ under screening. Left: positions with $\pm\Gamma/2$ bands; the uphill ($k = +1$) component crosses $E_r = 0$ at $\lambda \approx 0.11$ and continues as a positive-energy resonance. Right: widths.

splitting survives while the doublet as a whole rises. At $\lambda \approx 0.11$ the uphill component crosses $E_r = 0$. For $F \neq 0$ this is not a spectral threshold — the Stark Hamiltonian's spectrum covers the real line, and the scaled operator's eigenvalues continue analytically in the parameters [22] — so the smooth passage, confirmed numerically with no anomaly in position or width, is the expected analytic continuation. The physical content is that the state *retains* narrow, identifiable resonance character on the positive-energy side, reaching $E_r = +0.0837$, $\Gamma = 0.070$ at $\lambda = 0.30$: the screened-Stark analog of the positive-energy resonances of the field-free screened atom [16]. The downhill width grows from 0.061 to 0.160 over the same range, and the width ratio Γ_{-1}/Γ_{+1} drifts from 1.97 to 2.29. Coverage here is deliberately scoped: one field value, one manifold; the systematic excited-manifold survey is a natural continuation.

D. Physical scope

The computed ranges correspond to realizable plasma conditions: for electron densities 10^{21}–10^{22} cm^{-3} at temperatures of 1–2 eV, the electron Debye screening parameter is $\lambda \approx 0.2$–0.5 a.u. and the Holtsmark normal field $F_0 = 2.603\, e\, n_e^{2/3}$ is ≈ 0.007–0.034 a.u., so the grid covers the regime where the two perturbations genuinely co-occur. In a plasma the "static field" is of course the quasistatic ion microfield, a stochastic quantity distributed around F_0 [23]; the present uniform-field map supplies precisely the resonance parameters $E_r(F, \lambda)$, $\Gamma(F, \lambda)$ that enter a microfield average, which is the standard route from Stark parameters to line shapes.

IV. CONCLUSIONS

We have presented a coupled-channel complex-scaling method for atoms under combined Debye screening and a static electric field — validated to ten digits against the classical Stark benchmark, to 2.4×10^{-7} against an independent engine in the screening limit, and verified at run time against exact identities — and used it to compute what is, to our knowledge, the most extensive resonance map of the combined problem to date, together with the $n = 2$ doublet continued across $E = 0$. The map's physics organizes into a strongly asymmetric competition: screening amplifies field ionization exponentially (eight orders of magnitude at weak field), the spectroscopic-identity frontier recedes to weaker screening as the field grows, the perturbative Stark regime shrinks with screening until the width dominates the field response, and excited Stark components are pushed across the field-free threshold, persisting as positive-energy resonances.

The method extends without structural change to $m \neq 0$ sectors, higher manifolds, and other screening functions (only the diagonal potential blocks change); the exact-basis channels keep the matrices small enough that a full map is a matter of minutes on a laptop, which is what makes systematic coverage — rather than isolated points — routine. At mesh-level resolutions or for many-electron generalizations the same block structure is where contour-integral eigensolvers would become the natural tool. All scripts, certification batteries, and machine-readable grids are distributed with the companion programs and will be archived with a persistent identifier at publication.

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